

SVM(Support vector machine)

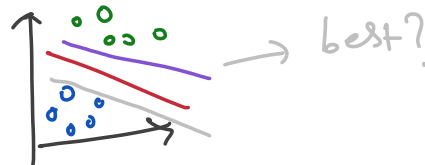
Sunday, August 09, 2026 7:55 PM

- It's a supervised learning algorithm.
- It's used for both classification & regression.
- Sometimes used also for anomaly detection.

Suppose we have two classes:

○ → class 1

● → class - 2



• Objective: To draw a line that separates them.

For 2 features, line is;

$$w_1x_1 + w_2x_2 + b = 0$$

or $w^T x + b = 0$ → compact form

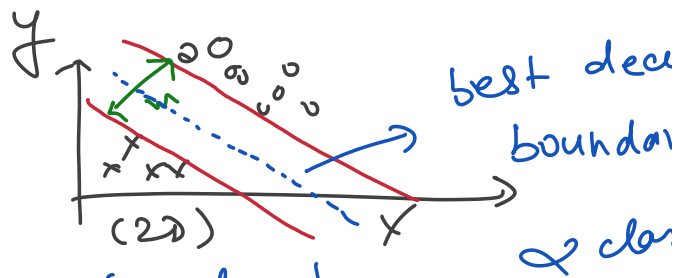
x → $n \times 1$ feature vector

b → bias

w → weight vector

Which separating line gives us the best margin betⁿ 2 classes.

SVM wants the **widest possible gap** between the classes



→ This is called as maximum margin classifier.

Hypersplane

In 2D, decision boundary is line.

$$w_1x_1 + w_2x_2 + b = 0$$

In 3D, it becomes a plane:

$$w_1x_1 + w_2x_2 + w_3x_3 + b = 0$$

In higher dimensions, it is called hyperplane.

$$w^T x + b = 0$$

Features	Decision boundary
2 features	Line
3 features	Plane
>3 features	Hyperplane

SVM Working:

Once SVM finds: $w^T x + b = 0$

We calculate $f(x) = w^T x + b$

$$\text{Then: } y = \begin{cases} +1 & w^T x + b > 0 \\ -1 & w^T x + b < 0 \end{cases}$$

positive $\rightarrow$ class 1
negative $\rightarrow$ class -1

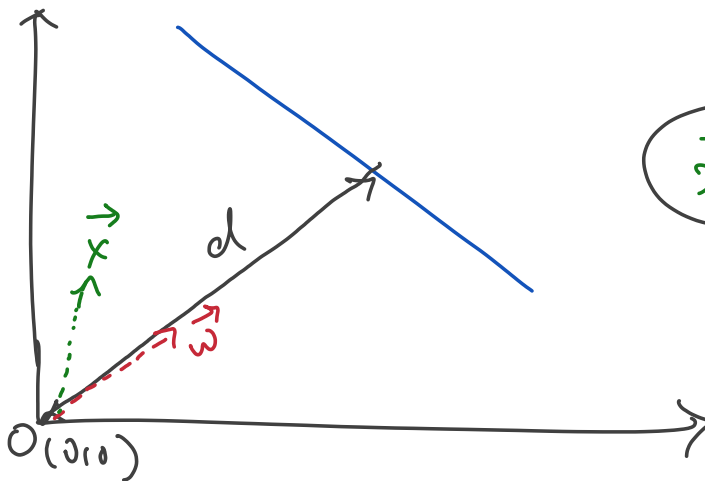
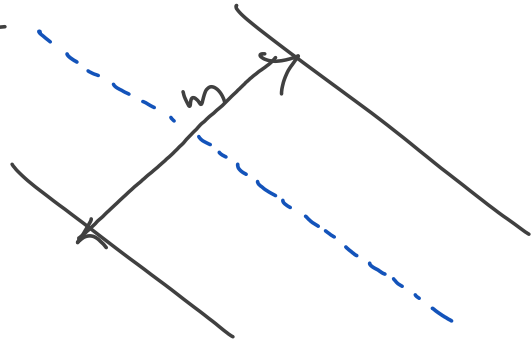
Margin

Our classifier is $w^T x + b = 0$

SVM would create additional parallel lines

$$w^T x + b = 1$$

$$w^T x + b = -1$$



Projection of

$$\frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|}$$

$$\frac{\vec{x} \cdot \vec{w}}{\|\vec{w}\|} = d$$

Numerical Example :-

say; $w = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

and $b = -5$

decision boundary: $2x_1 + x_2 - 5 = 0$

Now take a point: $x = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ or $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$x \cdot w = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 3 \cdot 2 + 2 \cdot 1 = 6 + 2 = 8$$

$$x \cdot w + b = 8 + (-5) = 3 > 0$$

point is class 1.

$$x \cdot w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= 2$$

$$x \cdot w + b =$$

$$=$$

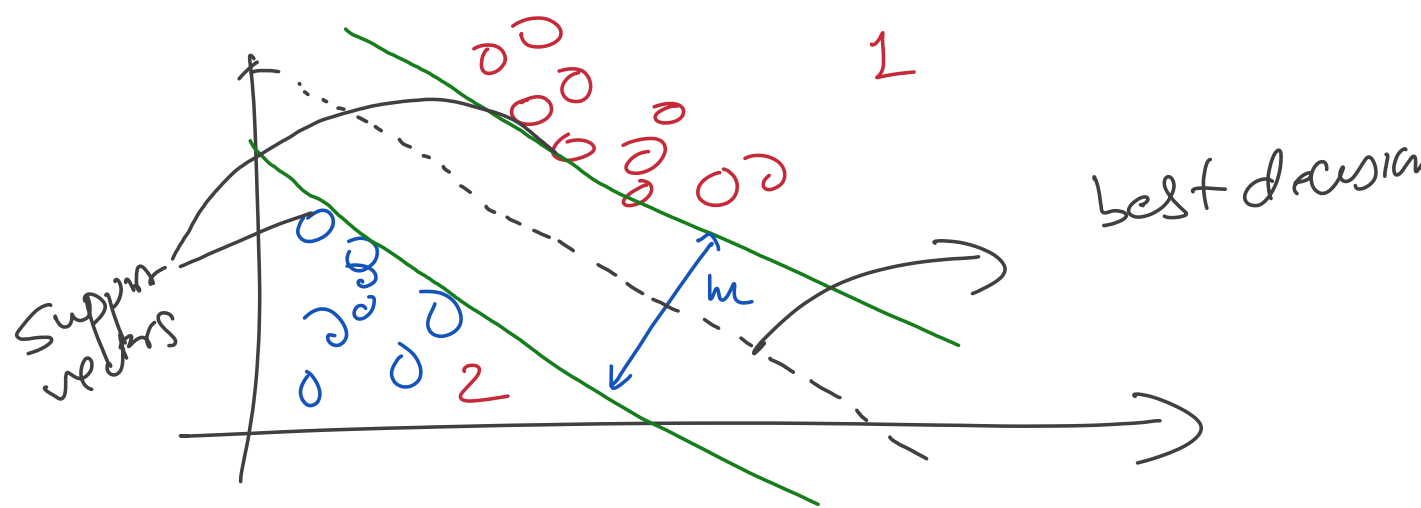
class - 1

$$\vec{x} \cdot \frac{\vec{w}}{\|\vec{w}\|} \geq d_2$$

$$\vec{x} \cdot \vec{w} \geq d_2 \cdot \|\vec{w}\|$$

$$\vec{x} \cdot \vec{w} - d_2 \|\vec{w}\| \geq 0$$

$$\vec{x} \cdot \vec{w} + b \geq 1 \quad (b = 1 - d_2 \|\vec{w}\|)$$



Derivation of margin

$$w^T x + b = 0 \text{ ————— decision boundary}$$

margin boundaries

$$w^T x + b = 1 \text{ ————— (1)}$$

$$w^T x + b = -1 \text{ ————— (2)}$$

Distance betⁿ 2 parallel hyperplanes:

$$\frac{|c_1 - c_2|}{\|w\|} \approx :$$

$$\text{Distance betⁿ (1) \& (2) = } \frac{2}{\|w\|}$$

$$\text{margin} = \frac{2}{\|w\|}$$